

NE523 Midterm Exam

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Part 1 Integro-differential Form

The one-speed, steady-state transport equation with a uniformly distributed monoenergetic isotropic source and vacuum boundary on square slab with side length a is

$$\left[\mu \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y} \right] \psi(x, y, \mu, \eta) + \sigma_t \psi(x, y, \mu, \eta) = S, \quad (1a)$$

$$x \in (0, a), \quad y \in (0, a), \quad \mu, \eta \in (-1, 1)$$

$$\psi(0, y, \mu, \eta) = 0, \quad \mu > 0 \quad (1b)$$

$$\psi(a, y, \mu, \eta) = 0, \quad \mu < 0 \quad (1c)$$

$$\psi(x, 0, \mu, \eta) = 0, \quad \eta > 0 \quad (1d)$$

$$\psi(x, a, \mu, \eta) = 0, \quad \eta < 0 \quad (1e)$$

where $\sigma_t(x, y) = \sigma_t$ is constant absorption cross section, and $S(x, y) = S$ is the uniformly distributed isotropic source (*neutrons cm⁻¹str⁻¹*).

Part 2 Integral Form

Since neutrons travel in a straight line (between collisions) in direction (μ, η) , any position in the neutron's path can be parameterized with

$$x = x_0 + \mu s \quad (2a)$$

$$y = y_0 + \eta s \quad (2b)$$

where s is the total distance traveled from the point (x_0, y_0) . Taking the derivative of ψ along the parameter s , using the chain rule, yields

$$\frac{\partial \psi}{\partial s} = \frac{\partial \psi}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial \psi}{\partial y} \frac{\partial y}{\partial s} \quad (3)$$

Using the derivative of Equation 2, this becomes

$$\frac{\partial \psi}{\partial s} = \frac{\partial \psi}{\partial x} \mu + \frac{\partial \psi}{\partial y} \eta, \quad (4)$$

which is equivalent to the bracketed term in Equation 1a. The transport equation can be converted into an ordinary differential equation by substituting Equations 2 and 4 into the integro-differential form:

$$\frac{d}{ds} \psi(x_0 + \mu s, y_0 + \eta s, \mu, \eta) + \sigma_t \psi(x_0 + \mu s, y_0 + \eta s, \mu, \eta) = S. \quad (5)$$

Part 3 Integration Along Characteristic

Integration of Equation 5 can be facilitated using an integrating factor

$$I(s) = \exp \left[\int_0^s \sigma_t ds'' \right] \quad (6)$$

where

$$\frac{d}{ds} I(s) = \sigma_t I(s). \quad (7)$$

Multiplying Equation 5 by $I(s)$ yields

$$\frac{d}{ds} \left\{ \psi(x_0 + \mu s, y_0 + \eta s, \mu, \eta) \right\} I(s) + \sigma_t I(s) \psi(x_0 + \mu s, y_0 + \eta s, \mu, \eta) = SI(s). \quad (8)$$

Using Equation 7, this can be rewritten as

$$\frac{d}{ds} \left\{ \psi(x_0 + \mu s, y_0 + \eta s, \mu, \eta) \right\} I(s) + \frac{d}{ds} \left\{ I(s) \right\} \psi(x_0 + \mu s, y_0 + \eta s, \mu, \eta) = SI(s), \quad (9)$$

which is, by inverting product rule for derivatives, equivalent to

$$\frac{d}{ds} \left\{ \psi(x_0 + \mu s, y_0 + \eta s, \mu, \eta) I(s) \right\} = SI(s). \quad (10)$$

Integrating this along the characteristic yields

$$\int_0^s ds' \psi(x_0 + \mu s', y_0 + \eta s', \mu, \eta) I(s') = S \int_0^s ds' I(s') \quad (11)$$

$$\psi(x_0 + \mu s', y_0 + \eta s', \mu, \eta) \exp \left(\int_0^{s'} \sigma_t ds'' \right) \Big|_{s'=0}^s = S \int_0^s ds' \exp \left(\int_0^{s'} \sigma_t ds'' \right). \quad (12)$$

Introducing optical length $\tau(x_0 + \mu s, y_0 + \eta s, x_0, y_0) = \int_0^s ds'' \sigma_t$, this can be written as

$$\begin{aligned} \psi(x_0 + \mu s', y_0 + \eta s', \mu, \eta) \exp[\tau(x_0 + \mu s', y_0 + \eta s', x_0, y_0)] \Big|_{s'=0}^s \\ = S \int_0^s ds' \exp[\tau(x_0 + \mu s', y_0 + \eta s', x_0, y_0)] \end{aligned} \quad (13)$$

$$\begin{aligned} \psi(x_0 + \mu s, y_0 + \eta s, \mu, \eta) \exp[\tau(x_0 + \mu s, y_0 + \eta s, x_0, y_0)] - \psi(x_0, y_0, \mu, \eta) \\ = S \int_0^s ds' \exp[\tau(x_0 + \mu s', y_0 + \eta s', x_0, y_0)] \end{aligned} \quad (14)$$

$$\begin{aligned} \psi(x_0 + \mu s, y_0 + \eta s, \mu, \eta) \exp[\tau(x_0 + \mu s, y_0 + \eta s, x_0, y_0)] \\ = \psi(x_0, y_0, \mu, \eta) + S \int_0^s ds' \exp[\tau(x_0 + \mu s', y_0 + \eta s', x_0, y_0)] \end{aligned} \quad (15)$$

$$\psi(x_0 + \mu s, y_0 + \eta s, \mu, \eta) = \frac{\psi(x_0, y_0, \mu, \eta) + S \int_0^s ds' \exp[\tau(x_0 + \mu s', y_0 + \eta s', x_0, y_0)]}{\exp[\tau(x_0 + \mu s, y_0 + \eta s, x_0, y_0)]} \quad (16)$$

3(a) Solution

Since σ_t is constant, the optical length is

$$\tau(s) = \int_0^s ds'' \sigma_t = \sigma_t s. \quad (17)$$

Since the boundary condition is vacuum, then $\psi(x_0, y_0, \mu, \eta) = 0$ for appropriate choice of (x_0, y_0) corresponding with the direction of the characteristic.

For $\eta > \mu > 0$, the solution is

$$\psi(s, \mu, \eta) = S \frac{\int_0^s ds' e^{\sigma_t s'}}{e^{\sigma_t s}} \quad (18)$$

$$= S \frac{\frac{1}{\sigma_t} [e^{\sigma_t s} - 1]}{e^{\sigma_t s}} \quad (19)$$

$$\psi(s, \mu, \eta) = \frac{S}{\sigma_t} [1 - e^{-\sigma_t s}] \quad (20)$$

where $s = \min\{\frac{x}{\mu}, \frac{y}{\eta}\}$ is the distance from the intersection of the characteristic curve and the boundary. Then the solution for $\eta > \mu > 0$ is

$$\psi(x, y, \mu, \eta) = \begin{cases} \frac{S}{\sigma_t} \left(1 - \exp \left[-\frac{\sigma_t}{\mu} x \right] \right), & \eta x < \mu y \\ \frac{S}{\sigma_t} \left(1 - \exp \left[-\frac{\sigma_t}{\eta} y \right] \right), & \eta x > \mu y \end{cases} \quad (21)$$

Part 4 Solution Sketch

Part 5 Other Cases

For other characteristic curves, the parameterized solution (Equation 20) does not change, only the definition of s .

For $\mu < 0, \eta < 0$, (x_0, y_0) now lies on the top or right boundaries ($x = a$ or $y = a$), so $s = \min \left\{ \frac{x-a}{\mu}, \frac{y-a}{\eta} \right\}$, and the solution becomes

$$\psi(x, y, \mu, \eta) = \begin{cases} \frac{S}{\sigma_t} \left(1 - \exp \left[-\frac{\sigma_t}{\mu} (x - a) \right] \right), & \eta(x - a) < \mu(y - a) \\ \frac{S}{\sigma_t} \left(1 - \exp \left[-\frac{\sigma_t}{\eta} (y - a) \right] \right), & \eta(x - a) > \mu(y - a). \end{cases} \quad (22)$$

Similarly, for $\mu < 0, \eta > 0$, (x_0, y_0) lies on the lower and right boundaries, $x = a$ and $y = 0$, so $s = \min \left\{ \frac{x-a}{\mu}, \frac{y}{\eta} \right\}$. The solution is

$$\psi(x, y, \mu, \eta) = \begin{cases} \frac{S}{\sigma_t} \left(1 - \exp \left[-\frac{\sigma_t}{\mu} (x - a) \right] \right), & \eta(x - a) < \mu y \\ \frac{S}{\sigma_t} \left(1 - \exp \left[-\frac{\sigma_t}{\eta} (y) \right] \right), & \eta(x - a) > \mu y. \end{cases} \quad (23)$$